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ORIGINAL PAPER

# Reduction in seroprevalence of viral transfusion-transmitted infections in southwest Nigeria in children with sickle cell disease using an enhanced screening strategy

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**Summary**

Blood transfusion is an integral component in the management of children and adults with sickle cell disease (SCD). Concerns about blood safety due to the high risk of bloodborne infections in sub-Saharan Africa limits the application of this cost-effective strategy in the management of individuals with SCD. In a single-centre, retrospective, longitudinal study in southwest Nigeria, we hypothesised that the use of stringent blood donor selection, along with very sensitive enzyme-linked immunosorbent assay (ELISA) screening methods would reduce transfusion-transmitted infections (TTIs). Among 45 002 eligible blood donors at the Lagos University Teaching Hospital in Nigeria, over a 5-year review period (2015–2019), the seroprevalence rate of viral TTIs was 9.83%. The seroprevalence rates for human immunodeficiency, hepatitis B, and hepatitis C viruses were 1.37%, 6.2%, and 2.25% respectively. Among 172 children with SCD, 71% (122/172) on regular blood transfusion and 29% (50/172) who had never been transfused or had less than two transfusions per lifetime, none acquired any TTIs using our enhanced screening approach during the study period. Thus, safe blood transfusion practices can be provided for children with SCD in sub-Saharan Africa with the use of stringent donor selection protocols and fourth-generation ELISA kits for TTI screening.

**KEYWORDS**
hepatitis B, hepatitis C, human immunodeficiency virus, Nigeria, regular blood transfusion, sickle cell disease, transfusion-transmitted infections

## INTRODUCTION

Sickle cell disease (SCD) affects 3.2 million people globally and ~176 000 people die of SCD-related complications per year.[1] Nigeria alone has been estimated to have at least 150 000 newborns with SCD annually, a prevalence of 30 per 1000 live births has been reported in a regional newborn screening programme.[2] SCD is an autosomal recessive, genetic disorder in which red blood cells (RBCs) sickle in deoxygenated states due to single nucleotide polymorphism in which GTG is substituted for GAG (thymidine for adenine) at the sixth codon of the β-globin chain. This leads to chronic haemolysis, varying degrees of anaemia, and vaso-occlusive complications.[3,4] Despite a 10-fold rise in erythropoiesis, the haemolysis and resultant anaemia is only partially compensated,[3] can be exacerbated by factors related and unrelated to

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SCD and has been associated with increased morbidity and premature death.[2,5–8]

Transfusion of RBCs remains a cornerstone in the management of this resultant anaemia in SCD. Regular blood transfusions are integral in the treatment and prevention of major SCD-related complications including strokes, acute chest syndrome, perioperative support, vaso-occlusive complications and acute splenic sequestration in children.[9–12] In Nigeria, 19.5%–57.5% of transfusions administered to children are used for the management of patients with severe SCD, a population often requiring repeated RBC transfusions.[13–16] However, the provision of safe and adequate blood supply remains a challenge in Nigeria and other countries in sub-Saharan Africa (SSA), mainly due to an elevated risk of viral transfusion-transmitted infections (TTIs).[17] The World Health Organisation (WHO) recommends mandatory screening of all blood donations for human immune deficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV) and syphilis. However, ~89% of 2.7 million units of whole blood collected from 40 countries in SSA in 2004 were reportedly not tested for HIV in a quality assured manner.[18] Many hospitals in this region still use rapid diagnostic test (RDT) kits to screen donor products for the major TTIs.[19,20] RDT kits have been shown to have lower sensitivities and specificities than enzyme-linked immunosorbent assay (ELISA).[21,22] ELISA-based serological testing for TTIs among donors has been proven to detect more TTIs, especially the fourth-generation ELISA kits.[23–25] The fourth-generation HIV kits are unique in that they can detect both the HIV antibodies and the HIV p24 antigen unlike the third-generation HIV kits that detect only antibodies to HIV. The approximate window periods for the available RDT are quite variable, but on average is 90 days. The window period for current third-generation tests is 22 days, while the window period for the fourth-generation tests is between 7–14 days.[26,27] This allows them to identify an infected individual earlier than the third-generation kits that identify only the HIV antibodies. TTI accounts for 5%–10% of HIV infection in SSA.[21] The prevalence of transfusion-transmitted HBV, HCV, syphilis and HIV among blood donors in Calabar, Nigeria was reported as 4.1%, 3.6%, 3.1% and 4.2% respectively.[28]

Nigeria has the second largest epidemic of HIV worldwide.[29] The current prevalence of HIV among youth and adults (aged 15–49 years) is 1.4%.[29] The pooled estimated prevalence of HBV in Nigeria is 13.6%.[30] This high burden of HIV and HBV infection in the population increases the challenge of providing a safe and adequate blood supply in Nigeria. The high prevalence of these viral infections is associated with an increased risk of donors being in a window period, in which they have been infected by a virus but do not yet have detectable virus-specific antibodies. The fourth-generation ELISA-based serological testing reduces the serological window period by 3–7 days before antibody detection.[31] In Nigeria, the serological method for mandatory blood donor screening for HBV, HCV and HIV is not standardised and differs across the nation, with some using the RDT method.[24,25]

We hypothesised that the use of a strict donor selection approach coupled with the use enhanced blood screening methodology would reduce the risk of TTI among chronically transfused children with SCD at the Lagos University Teaching Hospital (LUTH), Nigeria.

## PATIENTS AND METHODS

This was a single-centre, retrospective, longitudinal study conducted at the LUTH from 1 January 2015 to 31 December 2019. The LUTH is an 800-bed hospital with a blood donor centre that collects ~8000 units of whole blood annually. It is currently the major source of blood components in south-west Nigeria, producing an average of 900 units of RBCs per month, 450 whole blood-derived platelets units per month and 300 fresh-frozen plasma units per month. Paid donors are not accepted, only family replacement and voluntary donors are allowed to donate. Voluntary blood donors account for 12.3% of the blood donors and family replacement donors account for 87.7% of all donors.[32] All these blood components are produced from whole blood. About 80% of the blood donors are first-time donors. The hospital blood bank uses fourth-generation ELISA kits for screening of HIV and a third-generation kit for HBV and HCV. Manufacturers use the term 'generation' to show the development of the kits overtime in an effort to increase its specificity and sensitivity.[33] In HIV testing the fourth generation implies Ag–Ab-based ELISA detection, while earlier generations only detected antibodies. The third-generation HCV ELISA detect antibodies that bind to recombinant antigens derived from four HCV regions: core, non-structural 3, non-structural 4 and non-structural 5.

After obtaining Institutional Review Board (IRB) and ethics approval (ADM/DCST/HREC/APP/3165), the records of 45 002 blood donors were reviewed, and the results of the three routinely screened viral TTIs (HIV, HBV, HCV) were extracted. All donors were included in this study, and none were excluded. Blood donors at the blood bank are routinely screened using our enhanced hospital-specific flow chart (Figure 1). Potential donors who arrive at the donor clinic are screened verbally based on the blood bank questionnaire for risk of donating and risk of TTI. Our screening questionnaire is similar to those used in the USA, Canada, or Europe. Donors are asked about intravenous drug use, recent infection, hepatitis risk, men who have sex with men, but we do not ask about incarceration (Appendix S1). They are also examined for tattoos and intravenous drug use. All potential donors at high risk of TTI, including high-risk behaviours, tattoos or suspected intravenous drug users are excluded. In Nigeria, tattoos are usually performed under non-aseptic conditions, hence are a potentially high-risk factor for the transmission of HIV, HBV and HCV. Pre-donation HBV screening using DiaSpot rapid kits are also done because of high prevalence of HBV in the community. Donors

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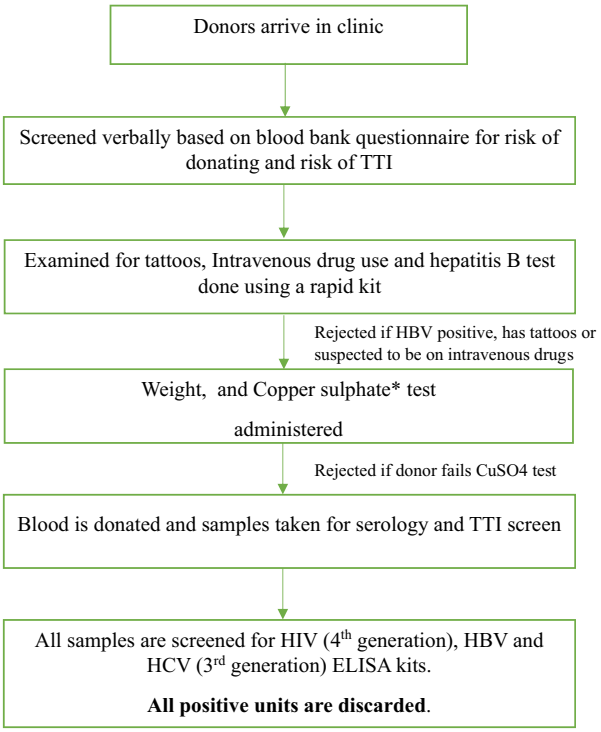



**FIGURE 1** Donor clinic algorithm flow chart for screening potential donor. *Copper sulphate test is a gravimetric method of approximating haemoglobin levels. ELISA, enzyme-linked immunosorbent assay; HCV, hepatitis C virus; HBV, hepatitis B virus; HIV, human immunodeficiency virus; TTI, transfusion-transmitted infection. [Colour figure can be viewed at wileyonlinelibrary.com]

haemoglobin levels are then estimated using the copper sulphate test.[34] We routinely measure the weights and heights of all donors. Pulse and blood pressure measurements are routinely taken during voluntary donor drives and in patients who give a history of hypertension. Temperature is not measured routinely during donor screening. Whole blood is then collected from donors who pass the initial screening steps. Samples for serological testing and TTI screening are also collected.

All donor samples are subsequently tested for HIV, HBsAg (hepatitis B surface antigen) and anti-HCV by ELISA techniques (Biokits: Bioelisa HIV 1+2 Ag/Ab, Barcelona, Spain; Bioelisa HBsAg 3.0, biokit, Barcelona Spain; Bioelisa HCV 4.0 respectively). Donated units of blood that are positive for any of the TTIs are discarded as per protocol. In addition, laboratory quality is assured by well-trained professionals. The Bioelisa HIV 1+2 Ag/Ab is an enzyme immunoassay (fourth generation) for the simultaneous qualitative detection antibodies to HIV-1 (groups M and O) or HIV-2 and HIV p24 antigen, with a sensitivity of 100% and a specificity of 99.76%–100%. The Bioelisa HBsAg 3.0 is a third-generation ELISA for qualitative detection of HBsAg in human serum or plasma with a sensitivity of 100% and specificity of 99.7%, while Bioelisa HCV 4.0 is also a third-generation enzyme immunoassay-based test for the detection of antibodies to HCV in human serum or plasma with sensitivity of 100% and specificity of 100%. About 5 ml of venous blood is withdrawn from all donors into sterile plain tubes for infectious disease screening. Sera were analysed with above reagents and all ELISAs were performed according to the manufacturers' instructions.

## Participants

A total of 172 children with SCD between the ages of 3–17 years who received care at the LUTH were included, 71% (122/172) were on regular blood transfusion therapy (manual exchange transfusion) as shown in Appendix S2 during the study period and 29% (50/172) had ≤2 units of RBC transfusion/lifetime or episodic transfusions. Patients on regular blood transfusion received manual exchange transfusion to achieve a HbS <30%, while those who received intermittent/episodic transfusions received simple transfusions. All RBC transfusions received must have come from the LUTH blood bank. HIV, HBV and HCV status pre-transfusion and at end of the transfusion programme for 122 children with SCD on regular blood transfusions during the study period (2015–2019) were obtained from case files. Those who had transfusions outside the hospital were excluded. HIV, HBV, and HCV status were also determined for 50 children with SCD who had had ≤2 units of RBC transfusions/lifetime. These patients were recruited from the paediatric SCD outpatient clinic.

As per institutional guidelines, all paediatric patients on regular blood transfusion are routinely screened for HIV, HBV, and HCV pre-transfusion, during and on completion of the transfusion programme. These results were obtained from their medical records. Similarly, the 50 children with SCD who had ≤2 units of RBC transfusion/lifetime were also screened for HIV, HBV and HCV. HBV and HCV status were determined using the DiaSpot test kit, while HIV status was determined by a serial HIV testing algorithm according to National Guidelines (Determine HIV 1/2 Ag/Ab test kit and Uni Gold HIV test kits). The DiaSpot test was not used for HIV, but rather for HBV (sensitivity 99%, specificity 97%) and HCV (Sensitivity 99.9%, specificity 98.6%) testing. These sensitivities are comparable to that of the Bioelisa ELISA kits used for blood donors. The Bioelisa HBsAg 3.0 has a sensitivity of 100% and specificity of 99.7%, while Bioelisa HCV 4.0 has a specificity and sensitivity of 100%. Transfusion history and biodata were obtained from case files.

## Statistics

Data were analysed using the IBM Statistical Package for the Social Sciences (SPSS®) for Windows, version 22 (IBM Corp., Armonk, NY, USA). Summary statistics such as frequencies and percentages were computed. Continuous variables were presented as means or medians depending on their distribution. Graphs were plotted on GraphPad Prism version 8.0.0 for Windows (GraphPad Software, www.graph

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pad.com). The number of blood donors who were positive per 100 donations were computed yearly from 2015 to 2019. Seroprevalence rates for the TTIs among children with SCD on regular blood transfusions and those not on regular transfusions were also calculated. Seroprevalence rates over the years were compared using chi-square statistics. Results were considered statistically significant at $p < 0.05$.

## RESULTS

A total of 45 002 blood donors were sampled over a 5-year period, retrospectively (2015–2019), with 9000 donors on average per year. The overall cumulative seroprevalence rate for TTI was 9.83%. HBV had the highest prevalence (6.2%) followed by HCV (2.25%), and HIV (1.37%). Table 1 shows a steady increase in numbers of donors over the review period. The infections showed significant inter-year variation ($p < 0.0001$). The prevalence of all three viral TTIs peaked in 2016, most notable was that of HBV, which showed a sharp decline after the year 2016. HIV and HCV showed a steadier decline with slight variations of <1.5% (Figure 2). The reason for this decline is not clear. A total of 172 children with SCD were enrolled, the mean (SD, range) age of the children was 7.46 (3.22, 2–18) years. Males accounted for 53.5% of the study participants (Table 2). The median (interquartile range [IQR]) number of RBC units transfused among those on regular transfusion was 16.5 (12–18) and 1 (0–1) among those on episodic transfusion during the entire study period. The maximum number of RBC units transfused among those on episodic transfusions was 2 units in 4% (two of 50); 1 unit in 60% (30/50) and 36% (18/50) had never been transfused. The median (IQR) duration of transfusion for those on regular transfusion was 8 (3–12) months (Table 3).

Primary stroke prevention was the main indication for regular blood transfusion in 90.1% (110/122); 73.4% (69/122) had been on regular transfusion for ≤12 months. Only two participants have been on regular transfusion for >36 months. All those on regular transfusions had manual exchange blood transfusion. About 80.3% (73/122) received an average of 1–2 units of packed RBCs per month. Routine iron overload screening showed 14/55 (25.5%) had serum ferritin values >1000 μg/L (Table 3). The median (IQR) serum ferritin level at the end of regular transfusion was 617.45 (412–1008) μg/L. Children with serum ferritin >1000 μg/L are routinely started on iron chelation with deferoxamine and/or deferasirox. All participants (regular transfusion and intermittent group) were seronegative for all routinely screened viral markers, which included HIV, HBV and HCV, at the end of the study period (Table 1).

## DISCUSSION

We have shown that the use of strict donor selection criteria coupled with the use of enhanced blood screening methodology, reduces the risk of TTI among transfused children with SCD in the LUTH, Nigeria. All children with SCD who received RBC transfusion during the study period (regular and episodic transfusion) remained seronegative for all routinely screened viral markers, including HIV, HBV and HCV. However, the provision of safe blood transfusion in SSA still faces numerous challenges such as the high prevalence of TTIs, financial constraints that limit the choice of donor-screening tests, lack of standardised hospital blood transfusion policies, and poor-quality assurance systems.[20,35–37]

The overall prevalence of TTIs among blood donors in Nigeria varied between 12.3% and 19.3% in previous studies, which is higher than the 9.83% reported in this study.[38–40] While blood donors in this study were screened for HIV using fourth-generation ELISA kits, most of the studies reported in Nigeria used RDT kits.[28,40,41] In a study by Fasola et al.[38] in Ibadan, Nigeria, a total of 41 445 blood donors were screened. The overall seroprevalence rate for TTIs was 12.3%. The prevalence was highest for HBV (8.5%) followed by HIV (1.8%), HCV (1.4%) and least for syphilis (0.5%), like our findings.[36] The highest TTI prevalence was reported in Kano, northern Nigeria (19.3%) in

**TABLE 1** Annual seroprevalence of HIV, HCV and HBV among donors and study participants during the study period

| Year | Total number of donors | HIV *n* (%) | HBV, *n* (%) | HCV, *n* (%) | Total with TTIs, *n* (%) | *p* |
|---|---|---|---|---|---|---|
| 2015 | 4840 | 73 (1.51) | 216 (4.46) | 79 (1.63) | 368.00 (7.63) | <0.0001 |
| 2016 | 7616 | 158 (2.07) | 714 (9.38) | 225 (2.95) | 1097.00 (14.40) | |
| 2017 | 11 213 | 117 (1.04) | 876 (7.81) | 226 (2.02) | 1219.00 (10.87) | |
| 2018 | 9732 | 115 (1.18) | 481 (4.94) | 268 (2.75) | 864.00 (8.88) | |
| 2019 | 11 601 | 154 (1.33) | 504 (4.34) | 216 (1.86) | 874.00 (7.53) | |
| Total | 45 002 | 617 (1.37) | 2791 (6.2) | 1014 (2.25) | 4422.00 (9.83) | |
| **Year** | **Total number of recipients** | **HIV, *n* (%)** | **HBV, *n* (%)** | **HCV, *n* (%)** | **Total with TTIs, *n* (%)** | ***p*** |
| 2015–2019[a] | 172 | 0 (0) | 0 (0) | 0 (0) | 0 (0) | N/A |

*Note*: Table shows a steady increase in numbers of donors over the review period.

Abbreviations: HBV, hepatitis B virus; HIV, human immunodeficiency virus; LUTH, Lagos University Teaching Hospital; TTIs, transfusion-transmitted infections.

[a]Recipients on study were screened routinely during study period pre and post.

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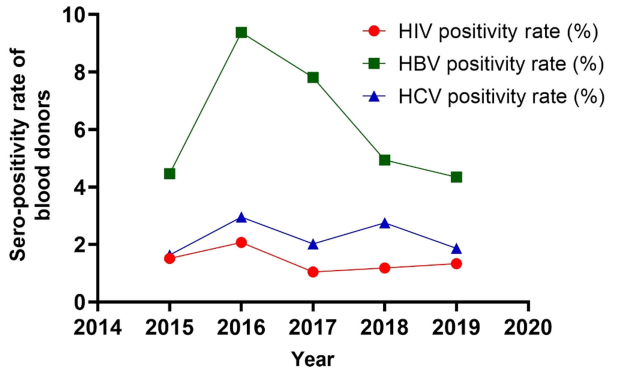



**FIGURE 2** Trend of seroprevalence of viral transfusion-transmitted infections among blood donors from 2015 to 2019 [Colour figure can be viewed at wileyonlinelibrary.com]

2012.[40] Similarly in Jos, a State in north central Nigeria, the overall TTI prevalence was 15.5% in 2015 using a third-generation ELISA kit.[39] This is twice the annual prevalence rate in 2015 among blood donors at the LUTH (Table 1), despite the fact that these two regions have comparable prevalence rates for HIV (Jos 1.5%; Lagos 1.3%).[42] The lower donor TTI prevalence in the LUTH is likely a result of our stringent blood donor selection criteria. We do not accept commercial donors, which are donors who receive money or other forms of payment for blood donation. First-time donors are recruited either through hospital-based donor drives or mobilised by family members, or community members of patients requiring blood transfusion. Compared to other African countries, the overall TTI rate in the LUTH is much lower than that reported in Kenya (14.1%)[43] and Sudan (20.1%),[44] but higher than that reported in Ethiopia (6.55%, 7.06%).[45,46]

The seroprevalence rate of HBV was highest in this study among donors (6.02%). Similar trends have been reported in several other studies in Nigeria[38–41,47,48] and other African countries.[44–46,49] Seroprevalence rates of HBV in the general population varies in the African region, ranging from 0.5% in Mauritius to 13.9% in Mali[50–53] and ranges in Nigeria have been reported between 4.3% and 23.3%.[54] Among blood donors in Nigeria, seroprevalence rates of HBV range between 2.0% and 10%.[28,41,47,55,56] This reflects the endemicity of HBV, high infectivity rate and its ability to survive for prolonged periods outside the body.[57,58]

Despite the fact that some studies in Africa have shown a higher prevalence of HBV and HCV among patients with SCD,[59–61] we did not corroborate this in our study. Studies in Nigeria which compared the seroprevalence rates of HBV and HCV among patients with SCD, with and without a

**TABLE 3** Clinical characteristics of participants with sickle cell disease on regular blood transfusion

| S/N | Characteristic | Value |
|---|---|---|
| 1. | Indication for transfusion ($n$ = 122), $n$ (%) | |
| | High-risk TCD | 109 (89.34) |
| | Stroke | 10 (8.4) |
| | TIA with silent cerebral infarcts | 1 (0.82) |
| | Recurrent persistent priapism | 1 (0.82) |
| 2. | Duration of regular transfusion in months ($n$ = 94), $n$ (%) | |
| | 0–12, $n$ (%) | 69 (73.4) |
| | 13–24, $n$ (%) | 22 (23.4) |
| | 25–36, $n$ (%) | 1 (1.1) |
| | >36, $n$ (%) | 2 (2.1) |
| | Duration of transfusion, months, median (IQR) | 8 (3–12) |
| 3. | Type of transfusion, $n$ (%) | |
| | Manual EBT | 122 (100) |
| 4. | Average number of packed RBC units/month ($n$ = 91), $n$ (%) | |
| | 1–2 | 73 (80.2) |
| | 3–4 | 14 (15.4) |
| | >4 | 4 (4.4) |
| 5. | Proportion on hydroxycarbamide ($n$ = 122), $n$ (%) | 120 (98.4) |
| 6. | Serum ferritin level ($n$ = 55), μg/L, median (IQR) | 617.45 (412–1008) |
| 7 | Proportion with serum ferritin >1000 μg/L ($n$ = 55), $n$ (%) | 14 (25.5) |

Abbreviations: EBT, exchange blood transfusion; IQR, interquartile range; SCD, sickle cell disease; TCD, transcranial Doppler; TIA, transient ischaemic attack.

**TABLE 2** Sociodemographic parameters of study participants

| Variable | Intermittent transfusion | Regular blood transfusions | All participants | $p$ |
|---|---|---|---|---|
| Age, years, mean (SD) | 8.32 (2.57) | 7.11 (3.39) | 7.46 (3.22) | 0.024[a] |
| Male, $n$ (%) | 28 (56.0) | 64 (52.5) | 92 (53.5) | 0.40[b] |
| Female, $n$ (%) | 22 (44.0) | 58 (47.5) | 80 (46.5) | |
| Manual exchange blood transfusion sessions, $n$, median (IQR) | NA | 6 (1) | | |
| RBC transfusion, units, median (IQR) | 1 (0–1) | 16.5 (12–18) | 12 (1–18) | <0.001[c] |

Abbreviations: IQR, interquartile range; RBC, red blood cell.

[a]Independent $t$-test.

[b]Fisher's exact.

[c]Mann–Whitney.

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history of transfusion with healthy age- and sex-matched controls, did not show a significant difference in prevalence rate of these viral markers among both groups.[62–64] However, a study from the northern part of Nigeria (Zaria) in 2002 reported one patient (one of 55; 1.8%) who was HIV-seropositive on chronic transfusion.[51] However, in the present study, none of the children with SCD on regular or episodic blood transfusion were seropositive for any of the viral TTIs (Table 1), even though the median number of RBC transfusions was 16.5 units for those on regular transfusions (Table 2). The median duration of regular transfusion was 8 months, and 26.6% of the chronically transfused children and adolescents in this study cohort had been on regular transfusion for >12 months.

Limitations of this study include the fact that we only tested for three viral TTIs (HIV, HBV, and HCV) that are considered mandatory for blood transfusion screening universally. Viral TTIs such as cytomegalovirus (CMV), which could present risk to immunocompromised recipients and other viral TTIs that are recommended only in specific countries with high endemicity (e.g., human T-cell lymphotropic [or leukaemia] viruses I/II), were not screened; thus, limiting the universal generalisability of this study, but our approach is clinically relevant in SSA. Selective screening is advised in these scenarios with up-front discussion with the transfusion medicine service. While we were diligent in screening health records of participants, we cannot completely exclude the possibility that transfusions among some participants may have occurred outside the LUTH in emergent situations.

In conclusion, blood transfusion represents an integral part of management of SCD, and an indispensable asset in low-resource settings. However, in low-resource settings where safe transfusion practices are limited, a strict donor selection criterion in combination with highly sensitive and cost-effective blood screening methods can help reduce the risk of TTI. This allows for safe and effective RBC transfusion therapy for the management of complications of SCD in SSA, a region endemic with the disease. To our knowledge, this is the first study from Nigeria and SSA showing that the use of strict donor selection criteria and use of enhanced blood screening methodology to reduce the risk of TTI among chronically transfused children with SCD. Fourth-generation ELISA tests are extremely sensitive and specific and they are more reliable in detecting early infection than rapid tests. They have median window period of 18 days for HIV, much lower than the third-generation kits, further reducing the residual risk of TTIs.[30–32] Implementation of more effective laboratory methods for TTI screening in local blood banks, intensifying voluntary blood donor motivation strategies and increasing community surveillance for TTI, will help to establish safer transfusion practices in SSA for transfusion-dependent populations.

## AUTHOR CONTRIBUTIONS

All authors contributed to the acquisition, analysis, and interpretation of the data; revised the manuscript critically; approved the final version for publication and agreed to be accountable for the results presented.

## ACKNOWLEDGEMENT

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## CONFLICT OF INTEREST

Authors have no conflict of interest.

## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

## PATIENT CONSENT STATEMENT

This was a retrospective longitudinal study conducted as part of standard care of patients after obtaining IRB and ethics approval. Written informed consent or verbal consent from the patient or patient's parent/guardian was not indicated.

## PERMISSION TO REPRODUCE MATERIAL FROM OTHER SOURCES

Not applicable in this case.

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